Where Is the Edge for FedAvg?

Edge of Stability in Federated Training

CS6007 Multi-Agent Machine Learning – Course Project Report

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Abstract

Full-batch gradient descent (GD) on neural networks trains at the *Edge of Stability* (EoS): the largest Hessian eigenvalue (the sharpness) rises until it reaches about $2/\eta$ and then stays there. Recent work reports the same behaviour in Federated Averaging (FedAvg), but does not explain where the federated edge lies or why the number of local steps K barely moves it. We linearize one FedAvg round into a *round map*, derive its stability condition, and test it in seven experiments on quadratics and on a tanh MLP trained on a CIFAR-10 subset with 10 clients. Four findings stand out. (i) In every federated run, the round map's growth per local step equals the EoS overshoot of plain GD, $c_{\text{eff}} = 1.077 \pm 0.010$ over 55 runs. (ii) With this one constant, the round map predicts the IID edge for all K and server step sizes η_s with a mean error of 1.9%, against 7.2% for the classical quadratic edge. (iii) Under non-IID data the clients stay at the same edge, and the global sharpness falls below $2/\eta$ by exactly a *misalignment* term M between the clients' sharp directions. M depends on the heterogeneity and hardly at all on K. (iv) The server step shifts the edge as predicted, including an even/odd effect in K.

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1 Problem statement and core idea

Notation. Throughout, $w \in \mathbb{R}^d$ are the model parameters and η is the local (client) step size. η_s is the server step size and K the number of local steps per round. There are N clients. Client i holds n_i samples, has data share $p_i = n_i / \sum_j n_j$ and local loss F_i. The global loss is

$$F(w) = \sum_{i=1}^{N} p_i F_i(w), \qquad H(w) = \nabla^2 F(w) = \sum_{i=1}^{N} p_i H_i(w), \quad H_i = \nabla^2 F_i. \tag{1}$$

The *sharpness* is $\lambda_{\max}(H)$, the largest eigenvalue of the Hessian. We always report sharpness normalized by the classical stability threshold,

$$S \;=\; \frac{\eta\, \lambda_{\max}}{2}, \tag{2}$$

so that $S = 1$ means "exactly at $2/\eta$". All values called "$\cdot\, \eta/2$" in tables and figures are of this form.

Edge of Stability in plain GD. For GD, $w^+ = w - \eta \nabla F(w)$. Near a point, a perturbation δ along a Hessian eigenvector with eigenvalue λ evolves as $\delta^+ = (1 - \eta\lambda)\, \delta$. It is stable iff $|1 - \eta\lambda| < 1$, that is $\lambda < 2/\eta$. Classical optimization theory therefore asks that sharpness stay below $2/\eta$. Cohen et al. [1] showed that neural networks do something else. Sharpness first grows (*progressive sharpening*), reaches $2/\eta$, and then hovers at or slightly above it while the loss keeps decreasing, non-monotonically. This regime is the Edge of Stability.

FedAvg. One round of FedAvg [7] starts every client at the server point w, runs K steps of local full-batch GD, and moves the server toward the average of the client endpoints:

$$w_i^{(0)} = w, \quad w_i^{(k+1)} = w_i^{(k)} - \eta \nabla F_i\big(w_i^{(k)}\big)\ (k < K), \qquad w^+ = w + \eta_s \sum_i p_i \big(w_i^{(K)} - w\big). \tag{3}$$

For $K = 1$ and $\eta_s = 1$ this is exactly one GD step on F.

The question. If FedAvg also trains at an edge, *where is that edge*? Is it $2/\eta$ in the global sharpness, the client sharpness, or something else? How does it depend on K, on data heterogeneity and on η_s?

Core idea: the round map. Freeze each client Hessian at the start of the round, $H_i = H_i(w)$. Then a perturbation δ of w passes through one round of (3) linearly:

$$\delta^+ = R\, \delta, \qquad R = I - \eta_s\, (I - A), \qquad A = \sum_i p_i\, (I - \eta H_i)^K. \tag{4}$$

In principle the round is stable iff the spectral radius $\rho(R) < 1$. On a real network $\rho(R) \gtrsim 1$ always holds, because flat directions ($H_i v \approx 0$) give $Rv \approx v$. We therefore track the complementary operator

$$T = I - R = \eta_s\, (I - A), \qquad \text{round stable} \iff \text{every eigenvalue } \tau \text{ of } T \text{ lies in } (0, 2). \tag{5}$$

For GD ($K = 1$, $\eta_s = 1$), $T = \eta H$ and $\tau_{\max}/2 = S$, so (5) contains the classical $2/\eta$ condition. The *growth per round* is

$$g = \max_{\tau} |1 - \tau| \quad \text{(over the extreme eigenvalues of } T), \tag{6}$$

and $g > 1$ means the frozen round is unstable. A round contains K local steps, so we also define the growth per local step, expressed on the same scale as S:

$$c_{\text{eff}} = \frac{1 + g^{1/K}}{2}. \tag{7}$$

For GD, $g = \eta\lambda_{\max} - 1$ at EoS and $c_{\text{eff}} = S$. In words, c_{eff} asks: if the round's instability were spread evenly over its K steps, how far past the edge would each step be?

Identical clients in closed form. If all $H_i = H$ with eigenvalue λ, then A has eigenvalue $a = (1 - \eta\lambda)^K$, and

$$\tau = \eta_s\big(1 - (1 - \eta\lambda)^K\big). \tag{8}$$

For $\eta_s = 1$, stability $|1 - \tau| = |1 - \eta\lambda|^K < 1$ is again $\lambda < 2/\eta$ *for every* K. This already suggests why K barely matters for IID data.

Our intention. We want to (a) check that the round map predicts the true divergence threshold where it should be exact (quadratics); (b) measure global sharpness, per-client sharpness and the round-map spectrum along real FedAvg training; and (c) test whether the federated edge is "the round map at its own edge", and how K, heterogeneity and η_s enter.

2 Prior work

- **Edge of Stability.** Cohen et al. [1] documented progressive sharpening and EoS for full-batch GD across architectures and losses. They also observed that with MSE loss the sharpness hovers slightly above $2/\eta$. Earlier, Jastrzebski et al. [5] found that the step size controls the sharpness reached early in training. Lewkowycz et al. [6] described the *catapult* phase: when training starts above $2/\eta$, the loss spikes and sharpness drops below the threshold.
- **Why EoS is stable.** Damian et al. [2] explain EoS through *self-stabilization*. Movement along the unstable direction, through the third derivative of the loss, pushes sharpness back down, so GD implicitly follows a constrained trajectory with $S \leq 1$. Arora et al. [3] and Ahn et al. [4] give related analyses of GD beyond $2/\eta$.
- **FedAvg and local steps.** FedAvg [7] and its analyses describe client drift under heterogeneous data. SCAFFOLD [8] corrects the drift with control variates, and FedProx [9] adds a proximal term. Local SGD theory [10] and the survey of Wang et al. [11] discuss when local steps help. Charles and Konečný [12] show that local update methods effectively minimize a surrogate objective that depends on η and K. Reddi et al. [13] make the server step a separate optimizer. Hsu et al. [14] introduced the Dirichlet(α) label split that we use.
- **Sharpness in federated learning.** Sharpness-aware FL methods [15, 16] *reduce* sharpness to improve generalization; they do not ask where the stability threshold lies.
- **EoS in FedAvg (closest work).** A recent preprint [17] reports that FedAvg also trains at EoS. Equilibrium sharpness scales as $1/\eta$, shifts with heterogeneity, and barely depends on K. The paper does not say why these hold, and does not predict the threshold.

3 What we want to improve on

The closest work reports *that* FedAvg sits at an edge; it does not say *which* edge, nor why K matters so little. Classical FL theory, in turn, assumes the step size is below $2/L$, which neural networks violate at EoS. We aim to:

1. give a **criterion** for the federated edge (the round map T), check it on models where it must be exact, and measure it on networks;

2. **explain the K-independence** (exact for identical clients, Eq. (8)) and show where and how K *does* matter;
3. separate the roles of **global** and **per-client** sharpness under heterogeneity;
4. predict the effect of the **server step** η_s, which the closest work did not study.

4 Hypotheses

These are the hypotheses as stated *before* the network experiments, based on the round map and the quadratic check.

Hypothesis 1 (self-tuning to the federated edge)**.** *FedAvg training self-tunes so that the frozen round map sits at its stability edge:* $\tau_{\max}/2 \approx 1$.

Hypothesis 2 (IID)**.** *With IID clients the global sharpness settles at* $\lambda_{\max}(H) \approx 2/\eta$*, i.e.* $S \approx 1$*, for every* K.

Hypothesis 3 (non-IID)**.** *With heterogeneous clients the global sharpness settles* below $2/\eta$*. The gap grows with heterogeneity, the sharpest client sits at its own edge,* $\max_i \eta\lambda_{\max}(H_i)/2 \approx 1$*, and the effect of* K *saturates as* K *grows.*

Hypothesis 4 (server step)**.** *At* $K = 1$ *the edge moves to* $\lambda_{\max}(H) \approx 2/(\eta\eta_s)$*. As* K *grows the effect of* η_s *fades, with an even/odd pattern in* K *predicted by Eq.* (8).

5 Experiments

This section only lists what was measured. Interpretation comes in Section 6. Observations are labelled (E$x.y$) for later reference.

5.1 Common setup

- **Data and model.** CIFAR-10 subset of $n = 5000$ training and 1000 test images, class balanced and standardized per channel. One-hot targets with MSE loss. A tanh MLP with two hidden layers of width 200 (about 6.6×10^5 parameters), float32 on GPU.
- **Clients.** $N = 10$ clients, labels split by Dirichlet(α) with $\alpha \in \{\infty, 1, 0.1\}$, where $\alpha = \infty$ means IID. The weighted mean total-variation distance between a client's label distribution and the global one is 0.053, 0.379 and 0.643 for the three splits. At $\alpha = 0.1$ the client sizes range from 132 to 893.
- **Training.** Full-batch local GD (Eq. (3)), all clients every round. Training length is counted in *local steps* (rounds $\times K$).
- **Measurements.** Hessian-vector products via forward-over-reverse automatic differentiation. Extreme eigenvalues come from Lanczos/ARPACK, warm-started. We record the global sharpness $\lambda_{\max}(H)$, every client's $\lambda_{\max}(H_i)$, the extreme eigenvalues $\tau_{\min}, \tau_{\max}$ of the frozen T (one matrix-vector product of T costs KN client Hessian-vector products), and the derived g and c_{eff}.
- **Equilibrium value.** The median over the last 30% of local steps.
- **Hardware and compute accounting.** RTX A6000 GPU, with jobs sharing a GPU. "GPU hours" below are the sum of per-run wall times.

5.2 Experiment 1: quadratic clients (round-map validation)

Setup. $N = 3$ clients with 2-D quadratic losses, equal weights and different minimizers. Client i has Hessian eigenvalues $10(1+hs_i)$ and 1, with $s_i \in \{-1, 0, 1\}$, rotated by $0.5\,hs_i$ radians. h controls heterogeneity; $h = 0$ means identical clients. **Sweep:** $K \in \{1, 2, 3, 5, 10, 20\}$, $h \in \{0, 0.3, 0.6, 0.9\}$, $\eta_s \in \{0.5, 1, 1.5\}$ (72 settings). For each setting, η runs over a 1000-point grid. We compare the smallest η at which the round map is unstable ($\rho(R) > 1$, exact here since there are no flat directions) with the smallest η at which a 3000-round float64 simulation diverges. Thresholds are reported as $\eta^\star\lambda/2$. **Compute:** CPU only, under a minute.

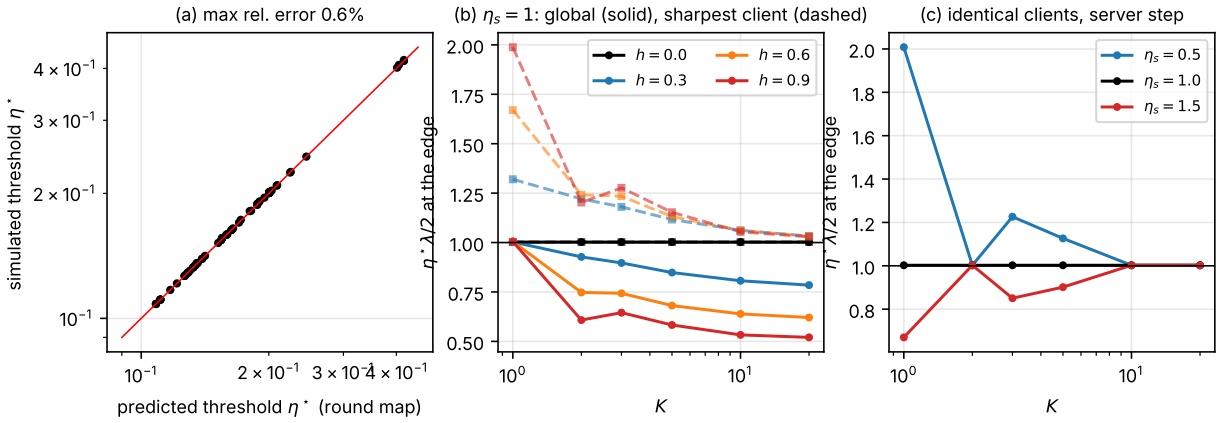


Figure 1: Experiment 1. (a) Round-map threshold vs simulated divergence threshold. (b) Edge in global (solid) and sharpest-client (dashed) normalized sharpness vs K, for $\eta_s = 1$. (c) Identical clients ($h = 0$) for three server steps.

Observations.

- **(E1.1)** Predicted and simulated thresholds agree in all 72 settings; the largest relative gap is 0.6%, one grid step.
- **(E1.2)** Identical clients, $\eta_s = 1$: the edge is $\eta^\star\lambda/2 = 1.00$ for every K.
- **(E1.3)** Heterogeneous clients, $\eta_s = 1$, $K = 1$: the global edge is still 1.00.
- **(E1.4)** For $K \geq 2$ the global value at the edge drops below 1. At $K = 2$ it is 0.93, 0.75 and 0.61 for $h = 0.3$, 0.6 and 0.9. For $h = 0.6$ the sequence over $K = 1, 2, 3, 5, 10, 20$ is $1.01, 0.75, 0.74, 0.68, 0.64, 0.62$, decreasing gradually up to $K \approx 10$.
- **(E1.5)** The sharpest client's value at the edge falls toward 1 as K grows: 1.03 at $K = 20$ for every h, against 1.32–1.99 at $K = 1$.
- **(E1.6)** Server step at $K = 1$: the edge is $1/\eta_s$, i.e. $\eta^\star\eta_s\lambda/2 = 1.00$ for $\eta_s = 0.5$ and 1.5.
- **(E1.7)** Server step with identical clients and $K > 1$: even K gives 1.00 for every η_s. Odd K does not. At $\eta_s = 0.5$, $K = 3$ and 5 give 1.23 and 1.13; at $\eta_s = 1.5$ they give 0.85 and 0.90. The odd-K deviation shrinks as K grows.

5.3 Experiment 2: centralized GD (EoS reproduction)

Setup. One client, $K = 1$, $\eta_s = 1$: plain full-batch GD on the 5k subset. **Sweep:** $\eta \in \{0.02, 0.04, 0.08\}$, 10^4 steps each, sharpness every 10 steps. **Compute:** 3 runs $\times \approx 40$ min $= 2.0$ GPU hours (all three shared one GPU).

Observations.

- **(E2.1)** Equilibrium sharpness scales as $1/\eta$: the ratios between consecutive η are 1.97 and 1.98.

η	$2/\eta$	equilibrium $\lambda_{\max}$	S	initial S	final loss
0.02	100	106.8	1.068	0.80	0.071
0.04	50	54.3	1.086	1.60	0.034
0.08	25	27.4	1.097	3.21	0.023

Table 1: Experiment 2: centralized GD equilibrium.

- **(E2.2)** The equilibrium sits 7–10% *above* $2/\eta$ ($S = 1.07$–1.10), increasing slightly with η, and drifts slowly down as the loss shrinks.
- **(E2.3)** Only $\eta = 0.02$ starts below the edge. There the sharpness dips, then rises (progressive sharpening) and crosses $S = 1$ near step 1200. At $\eta = 0.04$ and 0.08 training starts above the edge, and sharpness drops at once and settles, as in a catapult.

5.4 Experiment 3: single FedAvg run (timing and first look)

Setup. $\alpha = 0.1$, $K = 5$, $\eta = 0.02$, $\eta_s = 1$, 2000 local steps (400 rounds). **Compute:** 277 s (0.08 GPU hours). Of this, 73 s is training and 201 s (72%) is measurement, with 3068 matrix-vector products.

round	steps	global S	four sharpest clients, $\eta\lambda_{\max}(H_i)/2$	$\tau_{\min}$	$\tau_{\max}/2$
0	0	0.802	0.99, 0.90, 0.86, 0.85	−2.19	0.61
100	500	0.787	0.97, 0.89, 0.85, 0.83	−0.19	0.59
200	1000	0.974	1.16, 1.08, 1.03, 1.02	−0.21	1.21
300	1500	1.009	1.20, 1.15, 1.14, 1.07	−0.47	1.54
400	2000	1.016	1.20, 1.14, 1.12, 1.10	−1.32	1.58

Table 2: Experiment 3: one FedAvg run.

Observations.

- **(E3.1)** Global sharpness rises to $S \approx 1.01$–1.02 and stays there: FedAvg also shows EoS.
- **(E3.2)** The sharpest client stays at ≈ 1.20 from round 250 on, and the next three at 1.10–1.15.
- **(E3.3)** $\tau_{\max}/2$ reaches 1.58 (frozen round predicted unstable) while the loss keeps falling and training never diverges.
- **(E3.4)** $\tau_{\min}$ is already -2.19 at initialization, when all clients are below their edge.
- **(E3.5)** Measurement dominates the cost, mostly the round map, since each T-product costs KN Hessian-vector products.

5.5 Experiment 4: main FedAvg sweep

Setup. $\eta_s = 1$, 10^4 local steps per run. Global sharpness every 10 rounds; client sharpness and the frozen round map every 100 rounds. **Sweep:** $\alpha \in \{\infty, 1, 0.1\} \times K \in \{1, 2, 5, 10, 20\} \times \eta \in \{0.02, 0.04\}\times$ seeds $\{0, 1\}$, 60 runs, 4 jobs per GPU on 2 GPUs. **Compute:** 43.0 GPU hours of run time in total (6.1 h training, 36.8 h measurement). Training alone took ≈ 370 s per run for every K. One run ($\alpha = 1$, $K = 20$, $\eta = 0.04$) took 8.2 h because the eigensolver on T needed thousands of products. Five other runs at $K = 20$, $\eta = 0.04$ were stopped after about 8 h in their first round-map solve and rerun with the round map off; these aborted hours are not counted above.

Observations.

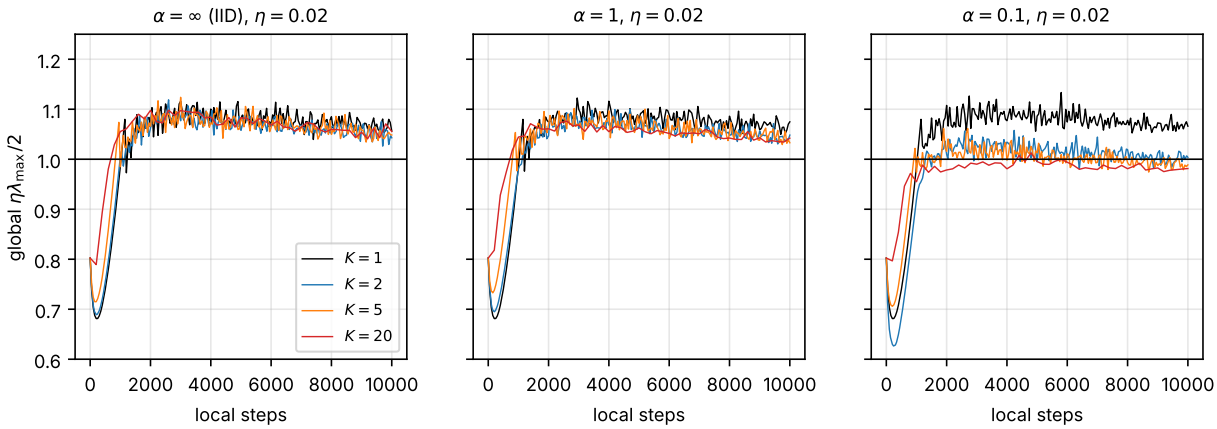


Figure 2: Experiment 4. Global normalized sharpness over training ($\eta = 0.02$, seed 0).

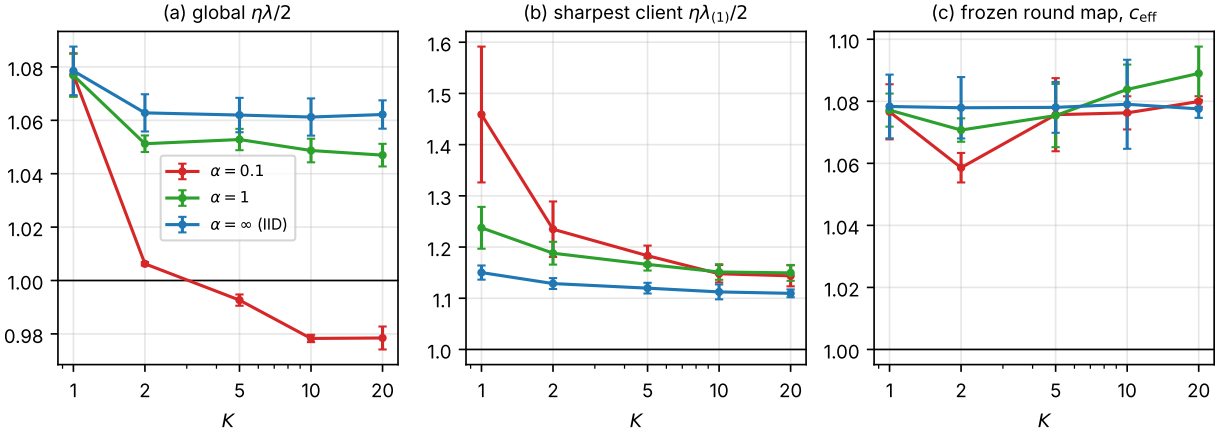


Figure 3: Experiment 4. Equilibrium values vs K, mean $\pm$ std over η and seeds. (a) Global S. (b) Sharpest client. (c) Per-step growth c_{eff} of the frozen round map.

- **(E4.1)** All 60 runs train stably; none diverges.
- **(E4.2)** Every run reaches EoS: global S rises (progressive sharpening at $\eta = 0.02$) and then hovers in a narrow band.
- **(E4.3)** IID: global S is flat in K for $K \geq 2$ (1.061–1.063) and slightly higher at $K = 1$ (1.079).
- **(E4.4)** At every $K \geq 2$, global S decreases with heterogeneity: IID $>$ $\alpha = 1$ $>$ $\alpha = 0.1$. At $\alpha = 0.1$ it is below 1 for $K \geq 5$ (0.993, 0.978, 0.978).
- **(E4.5)** At $K = 1$ global S is 1.077–1.079 for every α.
- **(E4.6)** At $\alpha = 0.1$ the drop from $K = 1$ to $K = 2$ is 0.071; further changes up to $K = 20$ total 0.028 and stop by $K = 10$. At $\alpha = 1$ and IID the change after $K = 2$ is at most 0.004.
- **(E4.7)** The sharpest client is always above its own edge. It is highest at $K = 1$ (1.46 at $\alpha = 0.1$), falls with K, and reaches 1.11–1.15 at $K = 20$ for all α.
- **(E4.8)** Scaling with η: global S is 0.01–0.015 higher at $\eta = 0.04$ than at 0.02 (IID: 1.057 vs 1.067 at $K \geq 2$); equilibrium sharpness scales as $1/\eta$ to within 2%.
- **(E4.9)** Seed-to-seed standard deviation of global S is ≤ 0.009.
- **(E4.10)** $\tau_{\max}/2$ zigzags with the parity of K: ≈ 1.5 at $K = 5$, ≤ 0.5 at even K. At even K, $\tau_{\min}$ becomes strongly negative and grows with K.

quantity	split	$K=1$	$K=2$	$K=5$	$K=10$	$K=20$
global S	$\alpha=\infty$	1.079	1.063	1.062	1.061	1.062
	$\alpha=1$	1.077	1.051	1.053	1.049	1.047
	$\alpha=0.1$	1.077	1.006	0.993	0.978	0.978
$\max_i \eta\lambda_{\max}(H_i)/2$	$\alpha=\infty$	1.150	1.129	1.120	1.112	1.110
	$\alpha=1$	1.238	1.188	1.166	1.151	1.150
	$\alpha=0.1$	1.459	1.235	1.183	1.148	1.144
c_{eff} (frozen)	$\alpha=\infty$	1.078	1.078	1.078	1.079	1.078
	$\alpha=1$	1.077	1.071	1.075	1.084	1.089
	$\alpha=0.1$	1.077	1.059	1.076	1.076	1.080
$\tau_{\max}/2$ (median)	all α	1.08	0.45–0.48	1.50–1.53	0.48–0.49	0.49
$\tau_{\min}$ (median)	all α	−0.09	−0.25 to −0.33	−0.40 to −0.53	−3.3 to −3.7	−17 to −24
g (median)	all α	1.15	1.25–1.33	2.0–2.1	4.3–4.7	18–25

Table 3: Experiment 4: equilibrium values (mean over two η and two seeds; the last three rows are medians over all runs of each K).

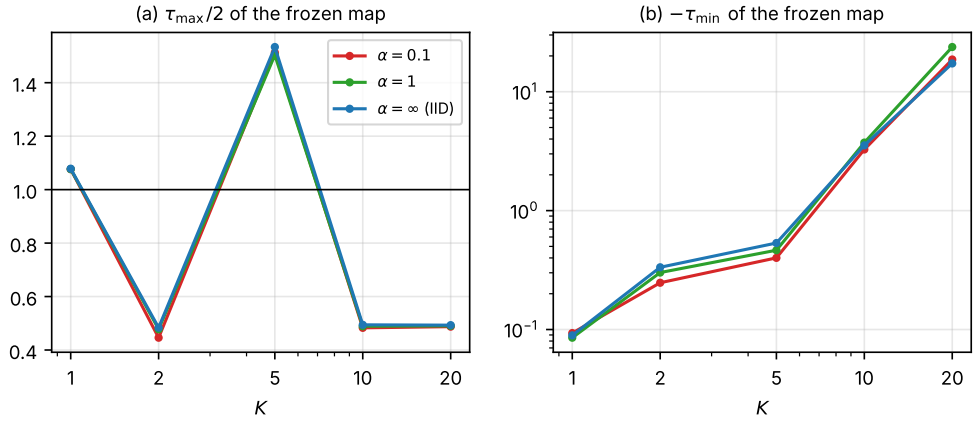


Figure 4: Experiment 4. Extreme eigenvalues of the frozen round map T at equilibrium vs K.

- **(E4.11)** The growth per round g grows quickly with K (1.15 at $K=1$ to 18–25 at $K=20$), while training is stable.
- **(E4.12)** The per-step growth c_{eff} is 1.06–1.09 in every setting; over 55 runs with a valid round map, $c_{\text{eff}} = 1.077 \pm 0.010$.
- **(E4.13)** At $\alpha=0.1$ the final training loss rises with K (0.055 at $K=1$ to 0.112 at $K=20$); test accuracy is 28–32% everywhere (the small MLP overfits the 5k subset).
- **(E4.14)** The cost of the round map grows with K, both per product (KN Hessian-vector products) and in the number of solver iterations at $K=20$.

5.6 Experiment 5: the real round vs the frozen round

Setup. The frozen map ignores that the Hessians change *during* the K local steps. We therefore also linearize the actual round: $J = \partial w^+/\partial w$, computed exactly by Jacobian-vector products through the K local steps, and $T_{\text{path}} = I - J$ (non-symmetric). We record both ends of its spectrum (smallest and largest real part) and the same g and c_{eff}. At $K=1$, $T_{\text{path}} = T$. Solver restarts are capped, and a growth value counts only if both ends converged. **Sweep:** $\alpha \in \{\infty, 0.1\} \times K \in \{1, 2, 5, 10\}$, $\eta = 0.02$, seed 0, 5000 steps, 6 round-map measurements per run (8 runs). **Compute:** 1.9 GPU hours.

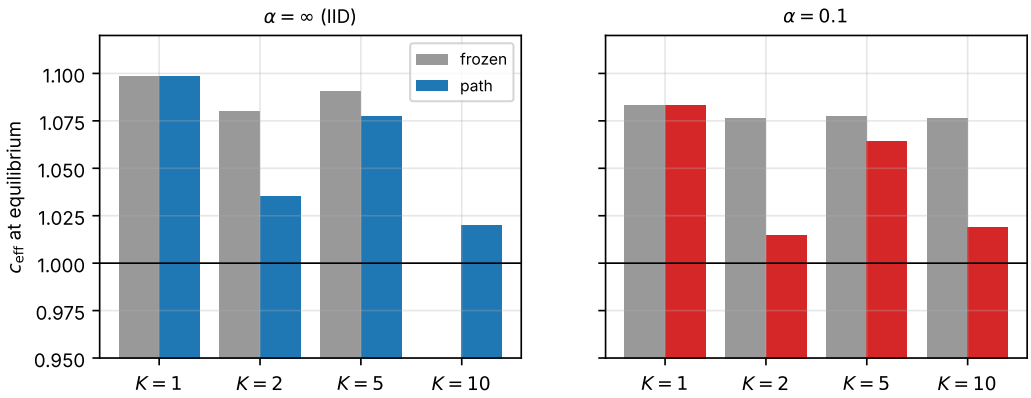


Figure 5: Experiment 5. Per-step growth at equilibrium for the frozen and the real (path) round map. The frozen value for IID, $K = 10$ did not converge.

split, K	global S	c_{eff} frozen	c_{eff} path	frozen $(\tau_{\min}, \tau_{\max})$	path (min, max real part)
IID, 1	1.085	1.099	1.099	$(-0.18, 2.18)$	$(-0.18, 2.18)$
IID, 2	1.072	1.080	1.035	$(-0.35, 0.98)$	$(-0.13, 0.98)$
IID, 5	1.077	1.091	1.077	$(-1.09, 3.30)$	$(-0.05, 3.06)$
IID, 10	1.081	—	1.020	—	$(-0.32, 1.32)$
$\alpha = 0.1$, 1	1.086	1.083	1.083	$(-0.13, 2.16)$	$(-0.13, 2.16)$
$\alpha = 0.1$, 2	1.029	1.077	1.015	$(-0.33, 0.89)$	$(-0.03, 0.90)$
$\alpha = 0.1$, 5	1.012	1.078	1.064	$(-0.48, 3.06)$	$(-0.03, 2.82)$
$\alpha = 0.1$, 10	1.003	1.076	1.019	$(-3.15, 0.98)$	$(-0.10, 1.00)$

Table 4: Experiment 5: equilibrium values.

Observations.

- **(E5.1)** At $K = 1$ the two maps agree exactly.
- **(E5.2)** For $K > 1$, path $c_{\text{eff}} \leq$ frozen c_{eff} in every case, and path $c_{\text{eff}} > 1$ in every case (range 1.015–1.077).
- **(E5.3)** Odd $K = 5$: path is only 0.013–0.014 below frozen.
- **(E5.4)** Even $K = 2, 10$: path is 1.015–1.035, against frozen ≈ 1.08.
- **(E5.5)** At even K the strongly negative frozen $\tau_{\min}$ almost vanishes along the path (to between -0.03 and -0.32). At odd K the large $\tau_{\max} > 2$ survives with a small reduction.
- **(E5.6)** With a 6-restart solver cap, 10 of 48 path and 8 of 48 frozen measurements failed to converge at one end of the spectrum. The frozen map at IID, $K = 10$ failed in 5 of 6 measurements.
- **(E5.7)** Global S at 5000 steps is slightly higher than the 10^4-step values of Experiment 4 (e.g. $\alpha = 0.1$, $K = 10$: 1.003 vs 0.978).

5.7 Experiment 6: alignment of client and global sharp directions

Setup. Let v be the top eigenvector of H. Since $H = \sum_i p_i H_i$ exactly,

$$\lambda_{\max}(H) = v^\top H v = \sum_i p_i \, v^\top H_i v \;\leq\; \sum_i p_i \, \lambda_{\max}(H_i) \;=:\; \bar{\lambda}, \tag{9}$$

where the inequality holds because $v^\top H_i v \le \lambda_{\max}(H_i)$ (Rayleigh, or Weyl's inequality [18]). The gap

$$M \;=\; \bar{\lambda} - \lambda_{\max}(H) \;=\; \sum_i p_i\big(\lambda_{\max}(H_i) - v^\top H_i v\big) \;\ge 0 \tag{10}$$

is the *misalignment cost*. It is zero iff every client is sharpest exactly along v. At the final point of each run we compute the top-5 eigenpairs of H and of every H_i, the Rayleigh quotients $v^\top H_i v$, and subspace overlaps $O(U,V) = \|U^\top V\|_F^2/5 \in [0,1]$ between top-5 eigenbases (robust to near-equal eigenvalues). **Sweep:** $K \in \{5,10\} \times \alpha \in \{\infty, 0.1\}$, $\eta = 0.02$, seed 0, 5000 steps (4 runs). **Compute:** 0.28 GPU hours; the alignment computation itself takes 12–14 s per run.

split, K	$\lambda_{\max}(H)$	$\bar{\lambda}$	M	sharpest client	overlap with global	pairwise overlap
IID, 5	1.082	1.103	0.021	1.137	0.89	0.81
IID, 10	1.075	1.095	0.020	1.127	0.87	0.79
$\alpha = 0.1$, 5	1.013	1.112	0.099	1.185	0.72	0.59
$\alpha = 0.1$, 10	0.998	1.085	0.088	1.146	0.78	0.62

Table 5: Experiment 6 (all eigenvalues $\cdot\, \eta/2$; overlaps are means over clients or client pairs).

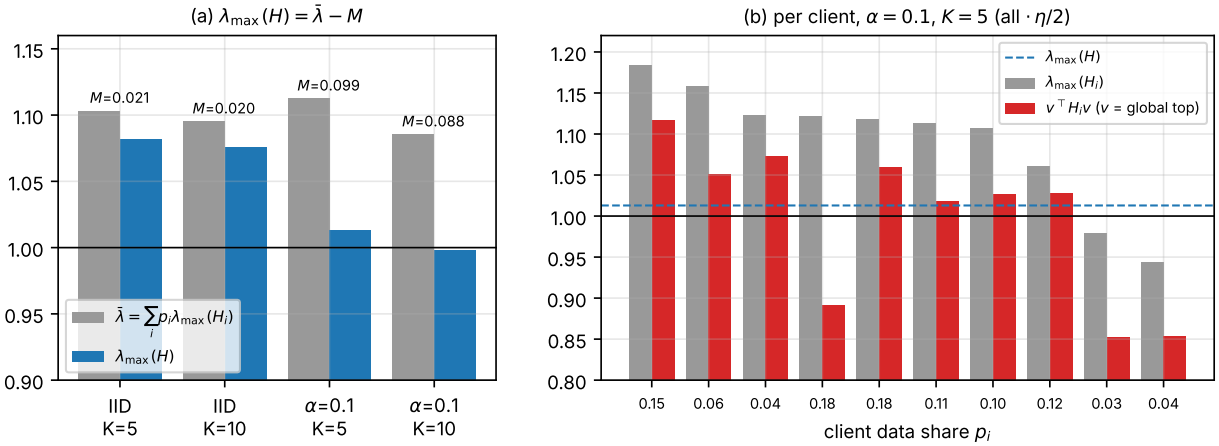


Figure 6: Experiment 6. (a) Decomposition (9). (b) Per client at $\alpha = 0.1$, $K = 5$: own top curvature vs curvature along the global top direction.

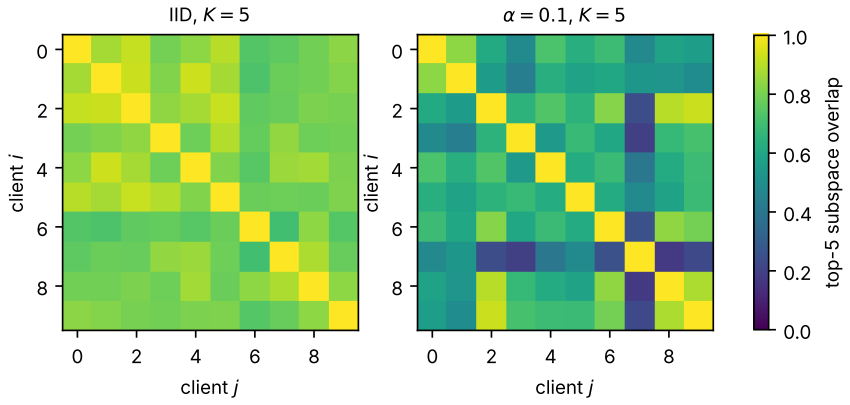


Figure 7: Experiment 6. Pairwise top-5 subspace overlap between clients.

Observations.

- **(E6.1)** The identity (9) holds numerically to a relative error of about 3×10^{-7} in all runs.

- **(E6.2)** The client average $\bar{\lambda}$ is ≈ 1.10 for both splits (1.085–1.112).
- **(E6.3)** M is 0.020–0.021 (IID) and 0.088–0.099 ($\alpha = 0.1$).
- **(E6.4)** Overlaps are lower at $\alpha = 0.1$: 0.72–0.78 with global (IID 0.87–0.89), 0.59–0.62 pairwise (IID 0.79–0.81).
- **(E6.5)** At $\alpha = 0.1$, one client (the largest, $p_i = 0.18$, labels almost only airplane, automobile and truck) contributes about 40% of M. Its overlap with the global subspace is 0.42–0.45, and it is about 20% flatter along v than along its own top direction.
- **(E6.6)** A client's label distance from the global label distribution correlates with its overlap with the global subspace ($r \approx -0.46$) and pairwise ($r \approx -0.43$); 10 clients only.
- **(E6.7)** The top two global eigenvalues are nearly equal ($\lambda_2/\lambda_1 = 0.97$–$0.995$), so single top eigenvectors are unstable even under IID; the top-5 subspaces are not.

5.8 Experiment 7: server step size

Setup. IID, $\eta = 0.02$, seed 0, 10^4 local steps, round map off; $\eta_s = 1$ is taken from Experiment 4. **Sweep:** $\eta_s \in \{0.5, 1.5, 2\} \times K \in \{1, 2, 5\}$ (9 runs). **Compute:** 1.4 GPU hours. The quadratic predictions for identical clients, from Eq. (8), are

$$S^\star = \begin{cases} 1/\eta_s & K = 1, \\ 1 & K \text{ even}, \\ \frac{1}{2}\left(1 + |1 - 2/\eta_s|^{1/K}\right) & K > 1 \text{ odd}. \end{cases} \tag{11}$$

Since networks overshoot ($S \approx 1.06$ at $\eta_s = 1$), we compare the *ratio* $S(\eta_s)/S(1)$ with $S^\star(\eta_s)$.

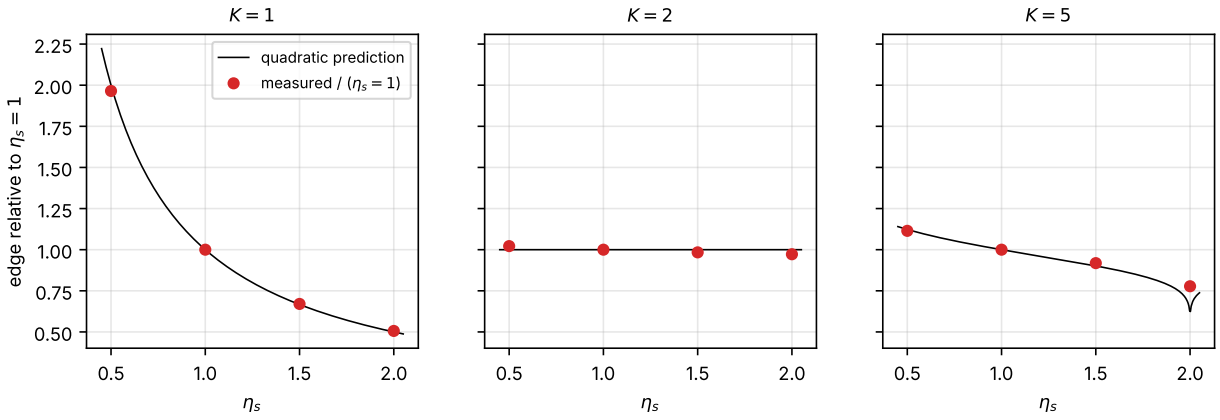


Figure 8: Experiment 7. Measured edge relative to $\eta_s = 1$ vs the quadratic prediction (11).

Observations.

- **(E7.1)** No run diverges.
- **(E7.2)** $K = 1$: measured ratios $1.965, 0.670, 0.506$ vs predicted $2, 0.667, 0.5$.
- **(E7.3)** $K = 2$: ratios $1.022, 0.984, 0.972$, near 1 for every η_s, with a slight downward trend in η_s.
- **(E7.4)** $K = 5$: ratios 1.115 and 0.918 vs predicted 1.123 and 0.901 at $\eta_s = 0.5$ and 1.5. At $\eta_s = 2$ the ratio is 0.778 vs predicted 0.5, and the prediction curve has a cusp there (Fig. 8).
- **(E7.5)** At $K = 1$ with $\eta_s \in \{1.5, 2\}$ training starts above the new edge (0.80 vs 0.67 and 0.5). Sharpness drops to ≈ 0.57 and ≈ 0.48 within the first 100 steps, then rises to the edge and stays.

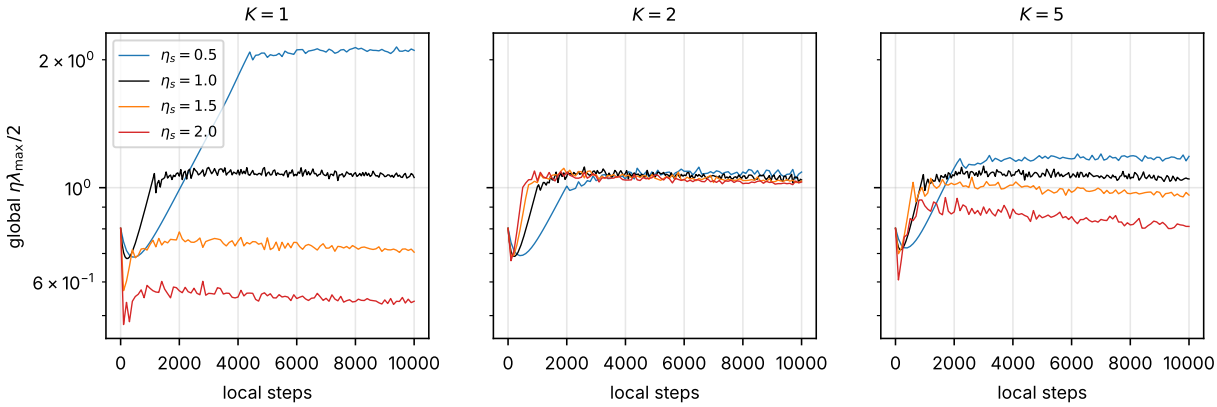


Figure 9: Experiment 7. Global normalized sharpness over training (log scale).

K	η_s	global S	sharpest client	ratio to $\eta_s = 1$	prediction (11)	final loss
1	0.5	2.105	2.247	1.965	2.000	0.106
1	1.5	0.718	0.766	0.670	0.667	0.045
1	2.0	0.542	0.580	0.506	0.500	0.039
2	0.5	1.081	1.159	1.022	1.000	0.141
2	1.5	1.041	1.102	0.984	1.000	0.028
2	2.0	1.029	1.096	0.972	1.000	0.032
5	0.5	1.178	1.246	1.115	1.123	0.111
5	1.5	0.970	1.025	0.918	0.901	0.055
5	2.0	0.822	0.864	0.778	0.500	0.072

Table 6: Experiment 7: equilibrium values. Reference values at $\eta_s = 1$: $S = 1.071, 1.058, 1.057$ for $K = 1, 2, 5$.

- **(E7.6)** With the per-step growth computed from Eq. (8) at the measured sharpness, $c_{\text{eff}} = 1.056 \pm 0.019$ over the 9 runs (1.04–1.08, except 1.02 at $K = 5$, $\eta_s = 2$).

5.9 Compute summary

6 Hypotheses: evidence and verdicts

6.1 H1: does training self-tune to the round map's edge?

In simple words. Plain GD keeps its sharpness at the point where one step is just barely unstable. H1 says FedAvg does the same with a whole *round*: the round map T should sit at its own edge, $\tau_{\max}/2 \approx 1$.

Evidence.

- On quadratics the round map is the exact stability criterion **(E1.1)**, so the criterion itself is right.
- On networks the raw round map is *not* at its edge. $\tau_{\max}/2$ reaches 1.5 at odd K **(E3.3, E4.10)**, and the growth per round is up to 25 at $K = 20$ **(E4.11)**, while every run is stable **(E4.1)**.
- The violation has a simple structure. In GD at EoS, every step multiplies the sharp direction by about $-(2S - 1) \approx -1.15$. The frozen map compounds this over K steps, so it predicts $g \approx 1.15^K$ per round: 2.0 at $K = 5$ and 16 at $K = 20$, close to the measured 2.0–2.1 and 18–25 **(E4.11)**.

#	experiment	runs	GPU hours (sum of run times)
1	quadratic clients	72 settings	CPU, < 0.02
2	centralized GD	3	2.0
3	single FedAvg run	1	0.08
4	main FedAvg sweep	60	43.0 (+ about 40 h of aborted solver time)
5	real vs frozen round	8	1.9
6	eigenvector alignment	4	0.3
7	server step	9	1.4
	total	85	48.7

Table 7: Compute. Jobs shared GPUs (2–5 per GPU), so device time was lower than this sum. Training was roughly 15–20% of it; the eigenvalue measurements were the rest.

- Undoing the compounding with Eq. (7) gives $c_{\text{eff}} = 1.077 \pm 0.010$ in all 55 runs, independent of α and K **(E4.12)**. This is the same band as plain GD's overshoot, 1.07–1.10 **(E2.2)**, and it rises with η as GD's does **(E4.8)**.
- The parity zigzag of $\tau_{\max}$ **(E4.10)** is expected. For even K, $(1-\eta\lambda)^K \geq 0$, so $\tau \leq \eta_s$ and the instability shows up instead as $\tau < 0$, i.e. $|1-\tau| > 1$ on the negative side. Using g over both ends of the spectrum removes the zigzag.
- Along the real trajectory **(E5.2–E5.5)**, the round is *at or slightly above* marginal stability per step ($c_{\text{eff}}^{\text{path}} \in [1.015, 1.077]$), never below. For odd K the per-step law holds almost unchanged **(E5.3)**. For even K the overshoot shrinks to about 0.3 of the frozen value **(E5.4)**, because the clients flatten within the round.

Why odd and even K differ (plausible mechanism). At EoS a client's iterate oscillates across the valley with period 2: the factor $1-\eta\lambda < -1$ flips its sign every step. After an odd number of steps the client ends the round on the opposite side, so the oscillation carries over to the next round. After an even number it returns to the same side, and the curvature change along the way cancels most of the compounding. The frozen map cannot see this, since it holds curvature fixed.

Verdict. Rejected as stated ($\tau_{\max}/2 \approx 1$ fails), **supported in reformulated form**:

$$\boxed{c_{\text{eff}} = \tfrac{1}{2}\left(1 + g^{1/K}\right) \;\approx\; c_{\text{GD}} \approx 1.08} \tag{12}$$

The frozen round map's growth *per local step* equals plain GD's EoS overshoot. Along the real round the overshoot is preserved for odd K and mostly removed for even K.

Caveats and open directions. At $K = 1$, $c_{\text{eff}} = S$ by construction, so only $K > 1$ is evidence. The reformulation was found after looking at the data; Experiments 5 and 7 are partly out-of-sample checks. Robustness tests: a third seed (planned), a second architecture or dataset, and the path map with more measurements per run (Experiment 5 has only 2–3 valid measurements in the equilibrium window).

6.2 H2: IID clients sit at $2/\eta$ for every K

In simple words. If all clients see statistically the same data, they have nearly the same Hessian. A round of K local steps is then like K steps of GD, and the edge should not move with K.

Mathematics. For identical clients and $\eta_s = 1$, Eq. (8) gives $|1-\tau| = |1-\eta\lambda|^K$. This exceeds 1 iff $|1-\eta\lambda| > 1$, i.e. iff $\lambda > 2/\eta$, for every K.

Evidence.

- Exact on quadratics: 1.00 for every K **(E1.2)**.
- On the network, IID global S is flat in $K \geq 2$ at 1.061–1.063 and 1.079 at $K = 1$ **(E4.3)**. All values are in GD's overshoot band **(E2.2)**, scale as $1/\eta$ **(E4.8)**, and have small seed variance **(E4.9)**.
- Even under IID the clients are not identical: $M_{\text{IID}} \approx 0.02$ **(E6.3)**, from finite samples (500 per client). This accounts for the small gap between $K = 1$ (global sharpness pinned, Section 7) and $K \geq 2$.

Verdict. Supported, with the edge at $c \cdot 2/\eta$, $c \approx 1.06$–1.08, not exactly $2/\eta$, like GD itself. This explains the SOTA observation that K barely matters under IID data.

Caveats. One architecture; IID splits with 500 samples per client.

6.3 H3: non-IID clients sit below $2/\eta$ [The Headline Result]

In simple words. With different data, clients are sharp in different directions. Each client can only be as sharp as its own stability allows. The global loss averages them, so where they disagree it is flatter than any single client.

Mathematics. Eq. (9)–(10): $\lambda_{\max}(H) = \bar{\lambda} - M$ with $M \geq 0$. If each client stays near its own edge ($\bar{\lambda} \approx c$), the global sharpness is $c - M$, below the GD level by exactly the misalignment.

Evidence.

- *Direction.* Global S decreases with heterogeneity at every $K \geq 2$ and drops below 1 at $\alpha = 0.1$ **(E4.4)**, as on quadratics **(E1.4)**.
- *Mechanism.* The decomposition is exact **(E6.1)**. The client average is the same for both splits (≈ 1.10, **(E6.2)**); what changes is M ($0.02 \to 0.09$–0.10, **(E6.3)**). The IID $-$ ($\alpha = 0.1$) gap in global sharpness (0.069 at $K = 5$, 0.077 at $K = 10$) equals the gap in M (0.078, 0.068) to within 0.01. Overlaps drop with heterogeneity **(E6.4)** and track label skew **(E6.5, E6.6)**.
- *Size.* Smaller than on quadratics: the network's global value is 0.98 at $\alpha = 0.1$, where quadratics gave 0.6–0.75 **(E1.4)**. The sharpest client is 11–15% above its edge, not at it **(E4.7)**. The network clients overlap more than the toy's rotated quadratics.
- *K dependence.* At $K = 1$ the global sharpness is pinned **(E4.5)**: one local step is a GD step on F, so heterogeneity cannot lower it. Clients then float far above their edge **(E4.7)**. For $K \geq 2$ most of the drop happens between $K = 1$ and 2, and it saturates by $K \approx 10$ **(E4.6)**. Quadratics show a more gradual decline **(E1.4)**.

Verdict. Supported qualitatively, refined quantitatively. Global sharpness sits below the GD level by the misalignment $M(\alpha)$, which grows with heterogeneity. "The sharpest client is at $2/\eta$" should be replaced by "the clients' *average* top sharpness is at the GD level c", and the sharpest client approaches ≈ 1.11–1.15 as K grows. The K effect saturates.

Caveats and open directions. One Dirichlet split per α, so the shape of $M(\alpha)$ cannot be fitted. M grows with η (0.105 at $\eta = 0.02$ vs 0.134 at 0.04, $\alpha = 0.1$). The $\alpha = 0.1$, $\eta = 0.04$, $K = 20$ run is still drifting at 10^4 steps. More splits, more clients, partial participation and other architectures would test robustness.

6.4 H4: the server step moves the edge

In simple words. The server step scales the whole round's update. At $K = 1$ it acts like a larger or smaller learning rate. For $K > 1$ the local steps already determine whether a client oscillates, so η_s matters less, and differently for odd and even K.

Mathematics. From Eq. (8) with $a = (1 - \eta\lambda)^K$: the round is unstable iff $\eta_s(1 - a) > 2$ or $\eta_s(1 - a) < 0$. For even K, $a \geq 0$, so the first condition never holds when $\eta_s \leq 2$, and the second holds iff $a > 1$, i.e. $\lambda > 2/\eta$, independent of η_s. For odd K instability needs $a < 1 - 2/\eta_s$, which gives Eq. (11). At $\eta_s = 2$, odd K, the condition is $a < 0$, i.e. $\eta\lambda > 1$: a cusp.

Evidence. Exact on quadratics **(E1.6, E1.7)**. On the network, 8 of 9 settings are within ≈ 2–3% of the prediction in ratio **(E7.2–E7.4)**. Training self-stabilizes through a catapult when it starts above the new edge **(E7.5)**. The one miss is at the cusp ($K = 5$, $\eta_s = 2$): there the per-round growth just above $\eta\lambda = 1$ is only $1 + 2(\eta\lambda - 1)^K$, about 1.2 per round even at $S = 0.82$. That instability is too weak to pull sharpness down to 0.5 **(E7.4, E7.6)**.

Verdict. Supported, including the even/odd structure; the network overshoots the predicted edge by the same factor as at $\eta_s = 1$.

Caveats. IID only, one seed, one η; the round map was off, so c_{eff} here is computed from the identical-client formula at the measured sharpness. Robustness: non-IID with $\eta_s \neq 1$, and a measured round map.

7 Empirical relations across all experiments

Pooling the experiments gives four relations. Values are $\cdot\, \eta/2$.

L1 (universal per-step overshoot). $c_{\text{eff}} = \frac{1}{2}(1 + g^{1/K}) = c$, with $c = 1.077 \pm 0.010$ (55 runs of Experiment 4), against $c_{\text{GD}} = 1.07$–1.10 (Experiment 2). The server-step runs give 1.056 ± 0.019.

L1a (IID edge in closed form). Substituting the universal growth into the identical-client map gives the edge as the smallest $x = \eta\lambda$ with

$$\left|1 - \eta_s\big(1 - (1 - x)^K\big)\right| = (2c - 1)^K. \tag{13}$$

Special cases: $\eta_s = 1 \Rightarrow S = c$ for every K; $K = 1 \Rightarrow S = c/\eta_s$; even $K \Rightarrow$ nearly independent of η_s. With one constant $c = 1.077$, this predicts the 29 IID runs (K from 1 to 20, η_s from 0.5 to 2) with a mean absolute error of 1.9%. The quadratic edge ($c = 1$) has an error of 7.2%. The largest errors are 14% at the cusp ($K = 5$, $\eta_s = 2$) and 6% at $K = 2$, $\eta_s = 0.5$.

L2 (non-IID decomposition). For every main-sweep run we computed $\bar{\lambda} = \sum_i p_i \lambda_{\max}(H_i)$ from the client sharpness and inferred $M = \bar{\lambda} - \lambda_{\max}(H)$. For $K \geq 2$:

$$\lambda_{\max}(H) \cdot \tfrac{\eta}{2} \;\approx\; \underbrace{c_0 + \frac{A(\alpha)}{K}}_{\bar{\lambda}\,\eta/2} - M(\alpha), \qquad c_0 \approx 1.085\text{–}1.087 \text{ for every } \alpha. \tag{14}$$

The client average tends to the same constant whatever the heterogeneity. M is nearly flat in K and set by the heterogeneity. The prediction reproduces the main sweep within ≈ 0.01, with the worst case 0.016 at $\alpha = 0.1$, $K = 2$. At $K = 1$ it is the global sharpness that is pinned at c, and the clients float above it ($\bar{\lambda} = 1.26$ at $\alpha = 0.1$). The direct measurement of Experiment 6 agrees ($M = 0.02$ and 0.09–0.10).

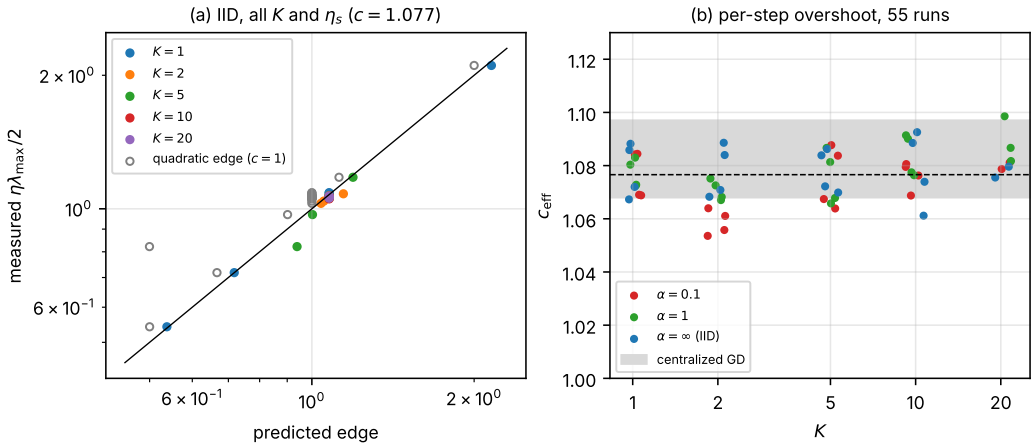


Figure 10: (a) L1a: predicted vs measured IID edge, all K and η_s. (b) L1: per-step overshoot of every main-sweep run against the centralized GD range.

split	c_0	A	M (mean over $K \geq 2$)
IID	1.086	0.012	0.026
$\alpha = 1$	1.087	0.007	0.038
$\alpha = 0.1$	1.085	0.113	0.120

Table 8: Fitted constants of L2.

L3 (sharpest client). $\max_i \lambda_{\max}(H_i) \cdot \eta/2 \approx c_{\max} + B(\alpha)/K$, with $R^2 = 0.97$–0.996 and $B = 0.04, 0.09, 0.33$ for IID, $\alpha = 1$ and $\alpha = 0.1$. A dilution argument explains the $1/K$. Within a round, the sharp direction of client i is amplified by roughly $w_i|1 - \eta\lambda_i|^K$, where $w_i \leq 1$ is that client's effective weight along its own direction. Setting this equal to the universal growth $(2c-1)^K$ gives

$$\frac{\eta\lambda_i}{2} \approx \frac{1 + (2c-1)\, w_i^{-1/K}}{2} \approx c + \frac{(2c-1)\ln(1/w_i)}{2K}. \tag{15}$$

Clients whose sharp directions are shared by few others have small w_i and may sit further above their edge, by an amount that decays as $1/K$.

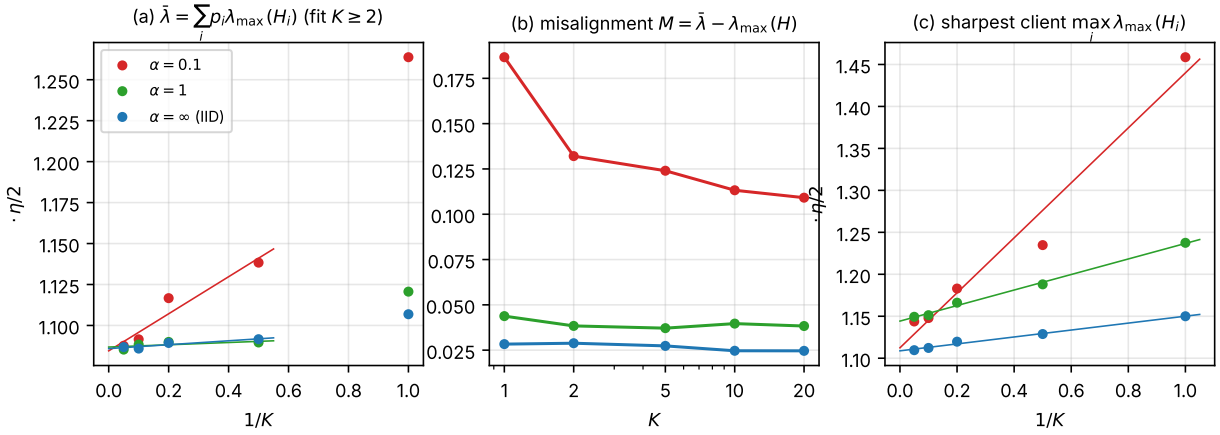


Figure 11: (a) L2: client-average top sharpness vs $1/K$ with fits for $K \geq 2$. (b) Misalignment vs K. (c) L3: sharpest client vs $1/K$ with linear fits.

L4 (parity along the real round). $c_{\text{eff}}^{\text{path}} - 1 \approx 0.9\,(c-1)$ for odd K and $\approx 0.3\,(c-1)$ for even K (Experiment 5; 8 runs, descriptive only).

Summary. FedAvg's edge is set by one number, GD's per-step EoS overshoot $c \approx 1.08$, pushed through the round map. Heterogeneity enters only through the misalignment M, which lowers the global sharpness while the clients stay at c.

8 Contributions relative to the state of the art

1. **A stability criterion for FedAvg at EoS.** The round map $T = \eta_s(I - \sum_i p_i(I - \eta H_i)^K)$ with $\text{spec}(T) \subset (0, 2)$ is exact on quadratics (0.6% error). We also show how it must be read on networks: per local step, using both ends of the spectrum because of the even/odd effect.
2. **An explanation of why K barely matters.** For IID data the edge is independent of K exactly (identical-client map). The SOTA observation follows, rather than being only reported [17].
3. **A universal constant.** The federated per-step overshoot equals GD's EoS overshoot (1.077 ± 0.010 across 55 settings). Together with the round map this gives a closed-form IID edge for any K and η_s, accurate to $\approx 2\%$.
4. **(Particularly interesting) The mechanism of the non-IID shift.** The exact decomposition $\lambda_{\max}(H) = \bar{\lambda} - M$ shows that clients stay at the GD level and that heterogeneity lowers the global sharpness only through misalignment of sharp directions. The direct eigenvector measurement confirms it. This answers "where is the edge" under heterogeneity: at the clients' average, not at the global loss.
5. **Where K does matter.** Only through heterogeneity, mostly between $K = 1$ and 2, saturating by $K \approx 10$. At $K = 1$ the global loss is pinned; for $K \geq 2$ the clients are, and the sharpest client approaches its limit as $1/K$.
6. **The server step**, which earlier EoS work on FL did not study. It rescales the edge as $1/\eta_s$ at $K = 1$, has no effect for even K, and follows a closed-form odd-K formula, all confirmed on the network.
7. **Real vs frozen round.** Clients partly self-stabilize within a round, strongly for even K and weakly for odd K. This is a new, testable parity effect.

9 AI Usage

- We used AI primarily for:
  - Code generation and debugging
  - Getting ideas on how to verify and consolidate the mathematical relations we observed being followed through the graphs of the experimental data
  - Editing report, where the report flow and format was dictated by us
- It also helped us decide which dataset and neural network to begin with based on its retrieval of the SOTA, as the work would be relevant only if we use similar setup.

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